

Docker Container Environment Setup in VS Code (Step-by-Step)

STEP 1 — Install Docker Desktop

Download Docker Desktop from <https://www.docker.com/products/docker-desktop> and install it with default settings.

STEP 2 — Fix WSL Update Error

Open PowerShell as Administrator and run:

```
wsl --update
```

After the update finishes, restart your computer.

STEP 3 — Verify WSL

Run:

```
wsl --version
```

You should see version details without errors.

STEP 4 — Open Docker Desktop

Open Docker Desktop. Wait until it shows 'Engine running'.

STEP 5 — Install VS Code

Download and install VS Code from <https://code.visualstudio.com>

STEP 6 — Install Docker Extension

Open VS Code.
Press Ctrl + Shift + X.
Search for 'Docker'.
Install the Docker extension by Microsoft.

STEP 7 — Create Project Folder

Create a folder named:

```
docker-project
```

STEP 8 — Open Folder in VS Code

In VS Code:
File → Open Folder → Select docker-project

STEP 9 — Create Files

Create these files inside the project:

```
app.py
requirements.txt
Dockerfile
```

STEP 10 — Write app.py

Paste this code:

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Docker Container Working"

app.run(host="0.0.0.0", port=5000)
```

STEP 11 — Write requirements.txt

Paste:

```
flask
```

STEP 12 — Write Dockerfile

Paste this code:

```
FROM python:3.10

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

CMD ["python", "app.py"]
```

STEP 13 — Open VS Code Terminal

Press:

```
Ctrl + `
```

STEP 14 — Build Docker Image

Run:

```
docker build -t myapp .
```

STEP 15 — Run Docker Container

Run:

```
docker run -p 5000:5000 myapp
```

STEP 16 — Open Browser

Open:

`http://localhost:5000`

You should see:

Docker Container Working

STEP 17 — Check Running Containers

Run:

`docker ps`

STEP 18 — Stop Container

Press Ctrl + C in terminal OR run:

`docker stop <container_id>`